

The minimum degree of binary polynomials vanishing to high order off the origin

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With substantial assistance from Claude and GPT models.

More details in Section [6.3](#).

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Abstract

Let $D(n, k)$ be the least degree of a polynomial $P \in \mathbb{F}_2[x_1, \dots, x_n]$ that vanishes with (Hasse) multiplicity at least k at every nonzero point of $\mathbb{F}_2^n$ and with multiplicity less than k at the origin. The multiplicity Schwartz–Zippel lemma of Dvir, Kopparty, Saraf and Sudan gives $D(n, k) \geq 2k - \lfloor k/2^{n-1} \rfloor$, a bound asked for by Bishnoi, Boyadzhiyska, Das and Mészáros, and their hyperplane covers attain it for $k \geq 2^{n-2}$. We determine $D(n, k)$ in the remaining range: for $1 \leq k < 2^{n-1}$,

$$D(n, k) = 2k + n - 2 - \lfloor \log_2 k \rfloor.$$

So $D(n, k)$ is known for all n and k . The Schwartz–Zippel bound is attained if and only if $k \geq 2^{n-2}$, and below 2^{n-2} each further dimension costs exactly one degree. The lower bound combines the multiplicity Schwartz–Zippel lemma with a one-dimension reduction that lowers the degree by one while keeping the origin order. The upper bound is a characteristic-two specialization of a Catalan truncation of Menezes, who determined the minimum for $n \geq k - 1$; a telescoping identity makes the construction work in every dimension. The main theorem and all its dependencies, including the multiplicity Schwartz–Zippel lemma and the hyperplane covers, are formalized in Lean 4, and every statement links to its formal proof.

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1 Introduction

Polynomials that vanish to high order on a grid are a basic tool of the polynomial method. Over the Boolean cube a natural extremal question asks how small the degree of such a polynomial can be when it is required to vanish to order k everywhere except at one point. Over $\mathbb{R}$ this question was answered by Sauermann and Wigderson [SW20]: the least degree of $P \in \mathbb{R}[x_1, \dots, x_n]$ with $P(\mathbf{0}) \neq 0$ and zeros of multiplicity at least $k \geq 2$ at all other points of $\{0, 1\}^n$ is $n + 2k - 3$. They observed that the answer depends on the field and that characteristic two behaves differently.

This paper determines the binary analogue. Throughout, multiplicity means Hasse multiplicity of formal polynomials (Definition 2.1); in particular no polynomial is reduced modulo $x_i^2 - x_i$, which would not preserve multiplicities.

Definition 1.1. For integers $n \geq 1$ and $k \geq 1$, a polynomial $P \in \mathbb{F}_2[x_1, \dots, x_n]$ is *k-admissible* if $\text{mult}_a(P) \geq k$ for every $a \in \mathbb{F}_2^n \setminus \{\mathbf{0}\}$ and $\text{mult}_{\mathbf{0}}(P) \leq k - 1$ [Lean]. Let $D(n, k)$ be the least total degree of a k -admissible polynomial.

The condition at the origin excludes $P = 0$ and allows any origin order below k , not only a nonzero value at the origin. Bishnoi, Boyadzhyska, Das and Mészáros [BBDM23, Question 4.3] asked, in the study of subspace coverings with multiplicities, whether $D(n, k) \geq 2k - \lfloor k/2^{n-1} \rfloor$ when $k \geq 2^{n-2}$. This bound follows for all n and k from the multiplicity Schwartz–Zippel lemma of Dvir, Kopparty, Saraf and Sudan [DKSS13]; see Proposition 3.1, which was the content of an earlier note [Note]. In the same paper they construct multisets of $2k - \lfloor k/2^{n-1} \rfloor$ affine hyperplanes that cover every nonzero point at least k times and the origin at most $k - 1$ times, for $k \geq 2^{n-2}$ [BBDM23, Theorem 1.2(a) of arXiv v1]. The product of the linear forms of such a cover is k -admissible, so the bound is attained for $k \geq 2^{n-2}$ (Theorem 5.3). This paper determines $D(n, k)$ in the remaining range.

1.1 Main result

Theorem 1.2. *For all integers $n \geq 1$ and $k \geq 1$,*

$$D(n, k) = \begin{cases} 2k + n - 2 - \lfloor \log_2 k \rfloor, & k < 2^{n-1}, \\ 2k - \lfloor k/2^{n-1} \rfloor, & k \geq 2^{n-1}, \end{cases}$$

and some k -admissible polynomial has degree exactly $2k - \lfloor k/2^{n-1} \rfloor$ if and only if $k \geq 2^{n-2}$ [Lean]. Moreover, for every $n \geq 1$ and $k \geq 1$ there is a k -admissible polynomial of degree at most $2k + n - 2 - \lfloor \log_2 k \rfloor$ [Lean].

The substance lies below 2^{n-1} , that is, for $n \geq \lfloor \log_2 k \rfloor + 2$ (Theorem 5.1). For $k \geq 2^{n-1}$ the value is the Schwartz–Zippel bound, attained by the hyperplane covers of [BBDM23] (Theorem 5.3). The two expressions agree for $2^{n-2} \leq k < 3 \cdot 2^{n-1}$ (Corollary 5.2), so the new values are those with $k < 2^{n-2}$. Three features of the formula stand out.

- **Where the Schwartz–Zippel bound is exact.** It is attained if and only if $k \geq 2^{n-2}$. For $k < 2^{n-2}$, $D(n, k)$ exceeds it by $n - 2 - \lfloor \log_2 k \rfloor \geq 1$.
- **Each dimension costs one degree.** For fixed k , $D(n, k) - n$ is constant for $n \geq \lfloor \log_2 k \rfloor + 2$. For example $D(n, 1) = n$, $D(n, 4) = n + 4$, $D(n, 8) = n + 11$ and $D(n, 15) = n + 25$.

- **Comparison with the reals.** Sauermann and Wigderson [SW20] show that over $\mathbb{R}$ the least degree with $P(\mathbf{0}) \neq 0$ is $n + 2k - 3$ for $k \geq 2$. Over $\mathbb{F}_2$ our construction with $P(\mathbf{0}) \neq 0$ (origin order $\ell = 0$) has degree $n + 2k - 2 - s_2(k - 1)$, which is $s_2(k - 1) - 1$ lower. Allowing any origin order below k lowers the degree further, to $D(n, k)$, which is $\lfloor \log_2 k \rfloor - 1$ below $n + 2k - 3$. Both savings are zero for $k = 2, 3$, and the second requires $n \geq \lfloor \log_2 k \rfloor + 2$.

We also prove the upper bound for each prescribed origin order in every dimension (Theorem 4.5): for $0 \leq \ell < k$ there is an admissible polynomial with origin order exactly ℓ and degree at most $n + 2k - 2 - s_2(k - \ell - 1)$, where s_2 is the binary digit sum.

1.2 Relation to previous work

- **Bishnoi, Boyadzhyska, Das and Mészáros [BBDM23].** They determine the least size of a multiset of affine hyperplanes (and, more generally, subspaces of codimension d) covering the nonzero points of $\mathbb{F}_2^n$ at least k times and the origin at most $k - 1$ times, for $k \geq 2^{n-2}$. Their covers give the upper bound for $k \geq 2^{n-2}$, and with Proposition 3.1, which answers their question on the polynomial version, they determine $D(n, k)$ there. We use their construction only for $k \geq 2^{n-1}$; for $2^{n-2} \leq k < 2^{n-1}$ ours has the same degree $2k$.
- **Menezes [Men26].** In a recent preprint, Menezes determines, over every field $\mathbb{F}$, the least degree $\delta_{\mathbb{F}}(n, k, \ell)$ of a polynomial with order at least k at the nonzero vertices of $\{0, 1\}^n$ and order exactly ℓ at the origin, for $k \geq 2$ and $n \geq k - 1$. In characteristic two his formula is $n + 2k - 2 - s_2(k - \ell - 1)$ [Men26, Corollary 1.3]. Minimizing over ℓ gives the value of $D(n, k)$ when $n \geq k - 1$. Together with the covers of [BBDM23], which settle $n \leq \lfloor \log_2 k \rfloor + 2$, our contribution is the range $\lfloor \log_2 k \rfloor + 3 \leq n < k - 1$, which grows with k . For $k = 2^j$ with $j \geq 3$, it is $j + 3 \leq n \leq 2^j - 2$, while $n \geq 2^j - 1$ is covered by [Men26]. The polynomials of our upper bound are Menezes' Catalan truncation specialized to $\mathbb{F}_2$. His degree formula is stated for $n \geq k - 1$, the range of his theorem. We read his statements, not his proofs, so we do not claim that his argument is limited to that range. The telescoping identity of Lemma 4.2 gives a short self-contained proof that the construction works in every dimension. Our lower bound is proved differently, by the one-dimension step of Theorem 3.5 from the multiplicity Schwartz–Zippel lemma.
- **Sauermann and Wigderson [SW20].** Their theorem is the real case described above. In characteristic two they give a polynomial of degree $n + 4$ for $k = 4$, below $n + 2k - 3 = n + 5$, showing that the real formula fails there.
- **Dvir, Kopparty, Saraf and Sudan [DKSS13].** Their multiplicity Schwartz–Zippel lemma is one of the two external inputs, with the hyperplane covers above. It gives the base case $n = \lfloor \log_2 k \rfloor + 2$ of the lower bound.
- **Alon and Füredi [AF93].** For $k = 1$, Theorem 1.2 says $D(n, 1) = n$ for $n \geq 2$. The lower bound is the case $\mathbb{F} = \mathbb{F}_2$ of their Theorem 5, which our argument reproves.

Two web searches, for high-order vanishing on the hypercube in positive characteristic and for punctured multiplicity Nullstellensätze over $\mathbb{F}_2$, found no statement for the range $n < k - 1$ other than the results above. A bounded search does not establish that none exists.

1.3 Proof outline

- **Lower bound (Section 3).** Take an admissible P in n variables. After an invertible linear change of coordinates in which the lowest homogeneous part at the origin is not divisible by x_n , the polynomial $S = P(x', 0) + P(x', 1)$ is admissible in $n - 1$ variables, has the same origin order, and has degree at least one lower (Theorem 3.5). Iterating down to $m = \lfloor \log_2 k \rfloor + 2$, where the multiplicity Schwartz–Zippel lemma already gives degree $2k$, yields $D(n, k) \geq 2k + n - m$.
- **Upper bound (Section 4).** With $y = x^2 + x$ we have $(1 + x) + \sum_{j < K} y^{2^j} = (1 + x)^{2^K}$ over $\mathbb{F}_2$. Expanding $\prod_i (1 + x_i)^{2^K}$ this way and discarding the terms of cost at least s leaves a polynomial g_s . It still vanishes to order s off the origin, takes the value 1 at the origin, and has small degree (Theorem 4.4). Multiplying by y_1^ℓ sets the origin order (Theorem 4.5).

1.4 Formal verification

Theorem 1.2, every lemma it uses, the multiplicity Schwartz–Zippel lemma and the hyperplane covers are proved in Lean 4 with Mathlib. The proofs use only Lean’s standard axioms. Each statement in this paper carries a [Lean] link to its formal proof at a fixed repository commit. Appendix A maps statements to declarations and gives the commands to check them. The two external inputs are stated without proof in the text, with citations, but their formal proofs are linked like every other statement, so the formal verification has no unchecked premise. Formal verification covers the statements under the linked definitions. It does not cover the prose, the comparison with the literature, or the fidelity of the definitions to the informal notions, although the definitions are short (Definitions 2.1 and 1.1).

2 Multiplicity and the Schwartz–Zippel lemma

Let $\mathbb{F}$ be a field and σ a finite set of variables. Degrees are total degrees of formal polynomials in $\mathbb{F}[x_\sigma]$. For a multi-index $\beta \in \mathbb{Z}_{\geq 0}^\sigma$ write $|\beta| = \sum_i \beta_i$.

Definition 2.1 (Hasse multiplicity). For $P \in \mathbb{F}[x_\sigma]$ and $a \in \mathbb{F}^\sigma$, write $P(a + z) = \sum_\beta c_\beta z^\beta$. The *multiplicity* of P at a is

$$\text{mult}_a(P) = \min\{|\beta| : c_\beta \neq 0\} \in \mathbb{Z}_{\geq 0} \cup \{\infty\},$$

with $\text{mult}_a(0) = \infty$ [Lean].

The coefficient c_β is the Hasse derivative $P^{(\beta)}(a)$, so this is the definition of Dvir, Kopparty, Saraf and Sudan [DKSS13, Definition 2.2]. In Lean, $\text{mult}_a(P)$ is the order, in $\mathbb{N}_\infty$, of the power series of the shifted polynomial $P(a + z)$.

Lemma 2.2. Let $P, Q \in \mathbb{F}[x_\sigma]$ and $a \in \mathbb{F}^\sigma$.

1. For $m \in \mathbb{Z}_{\geq 0}$, $\text{mult}_a(P) \geq m$ if and only if every coefficient of $P(a + z)$ of degree less than m vanishes.
2. $\text{mult}_a(P + Q) \geq \min(\text{mult}_a(P), \text{mult}_a(Q))$.
3. $\text{mult}_a(PQ) = \text{mult}_a(P) + \text{mult}_a(Q)$. (The inequality $\geq$ holds over every commutative ring, equality over every integral domain.)

4. $\text{mult}_a(P) = 0$ if and only if $P(a) \neq 0$.
5. If $e : \sigma \rightarrow \tau$ is a bijection and $P^e \in \mathbb{F}[x_\tau]$ is obtained by renaming x_i to $x_{e(i)}$, then $\text{mult}_b(P^e) = \text{mult}_{b \circ e}(P)$ for all $b \in \mathbb{F}^\tau$.

[Lean]

Proof. Part 1 is the definition. Part 2 holds because a coefficient of the sum of degree below both minima is a sum of two zero coefficients. For part 3, the lowest homogeneous parts of $P(a+z)$ and $Q(a+z)$ multiply to the part of degree $\text{mult}_a(P) + \text{mult}_a(Q)$ of $(PQ)(a+z)$; over a field this product is nonzero when both factors are. Part 4 holds because the constant coefficient of $P(a+z)$ is $P(a)$. For part 5, renaming commutes with the shift, and it maps the monomials of degree d bijectively to the monomials of degree d . $\square$

The only result we use without proof is the multiplicity Schwartz–Zippel lemma.

Theorem 2.3 (Multiplicity Schwartz–Zippel [DKSS13, Lemma 2.7]). *Let $\mathbb{F}$ be a field, σ a nonempty finite set, $P \in \mathbb{F}[x_\sigma]$ a nonzero polynomial and $S \subseteq \mathbb{F}$ a finite set. Then*

$$\sum_{a \in S^\sigma} \text{mult}_a(P) \leq \deg P \cdot |S|^{|\sigma|-1}$$

[Lean].

The Lean proof follows the induction on the number of variables in [DKSS13] (Lemma 8 of the arXiv version). It also proves the scaled form $|S| \sum_a \text{mult}_a(P) \leq \deg P \cdot |S|^{|\sigma|}$, which holds with no variables.

3 The lower bound

From now on the field is $\mathbb{F}_2$ unless stated otherwise.

3.1 The Schwartz–Zippel bound

Proposition 3.1. *Let $n \geq 1$ and $k \geq 0$. Every nonzero $P \in \mathbb{F}_2[x_1, \dots, x_n]$ with $\text{mult}_a(P) \geq k$ for all $a \neq \mathbf{0}$ satisfies*

$$\deg P \geq 2k - \lfloor k/2^{n-1} \rfloor$$

[Lean]. *In particular this holds for every k -admissible P .*

Proof. Put $d = \deg P$ and $m = 2^{n-1}$. The $2^n - 1$ nonzero points contribute at least $k(2^n - 1)$ to $\sum_{a \in \mathbb{F}_2^n} \text{mult}_a(P)$, and Theorem 2.3 with $S = \mathbb{F}_2$ bounds this sum by dm . So $k(2m - 1) \leq dm$.

Let $q = \lfloor k/m \rfloor$, and suppose $d + q < 2k$. Then $dm + (q + 1)m \leq 2km$. Since $(q + 1)m > k$, this gives $dm < 2km - k = k(2m - 1)$, a contradiction. $\square$

This is the deduction of the earlier note [Note], stated without the restriction $k \geq 2^{n-2}$ of [BBDM23, Question 4.3]. As the note observed, the origin condition is used only to exclude $P = 0$.

3.2 Linear substitutions

For a matrix $A \in \mathbb{F}^{\sigma \times \sigma}$ over any field, let $P \mapsto P \circ A$ be the algebra map with $x_i \mapsto \sum_j A_{ij}x_j$. The same matrix acts on points by $(Aa)_i = \sum_j A_{ij}a_j$.

Lemma 3.2. *Let $A, B \in \mathbb{F}^{\sigma \times \sigma}$ and $P \in \mathbb{F}[x_\sigma]$.*

1. *If P is homogeneous of degree d , so is $P \circ A$. Hence $P \mapsto P \circ A$ commutes with taking homogeneous components, and $\deg(P \circ A) \leq \deg P$.*
2. *$(P \circ A) \circ B = P \circ (AB)$.*
3. *If A is invertible, then $\text{mult}_0(P \circ A) = \text{mult}_0(P)$ and $\text{mult}_a(P \circ A) = \text{mult}_{Aa}(P)$ for every a .*

[Lean]

Proof. Part 1 holds because each x_i goes to a linear form. Part 2 can be checked on the variables. For part 3, by part 2 an invertible A gives an injective substitution. The origin order is the least j with a nonzero homogeneous component of degree j , and by part 1 this is preserved.

For the second claim, the shift commutes with substitution: $(P \circ A)(a + z) = P(Aa + Az) = (P(Aa + z)) \circ A$. So $\text{mult}_a(P \circ A) = \text{mult}_0(P(Aa + z) \circ A) = \text{mult}_0(P(Aa + z)) = \text{mult}_{Aa}(P)$. $\square$

Lemma 3.3. *Write $\lambda_v = \sum_i v_i x_i$ for the linear form of a vector v .*

1. *Let $L \in \mathbb{F}_2[x_\sigma]$ be nonzero with $\deg L < 2^{|\sigma|} - 1$. Then $\lambda_v \nmid L$ for some nonzero $v \in \mathbb{F}_2^\sigma$.*
2. *Over any field $\mathbb{F}$, for every nonzero $v \in \mathbb{F}^\sigma$ and every $i_0 \in \sigma$ there is an invertible A with $\lambda_v \circ A = x_{i_0}$.*

[Lean]

Proof. (1) The ring $\mathbb{F}_2[x_\sigma]$ is a unique factorization domain. Each nonzero λ_v has degree one, so it is irreducible, hence prime. The only unit of $\mathbb{F}_2[x_\sigma]$ is 1, so distinct forms are not associated. Distinct vectors give distinct forms. If all $2^{|\sigma|} - 1$ nonzero forms divided L , their product would divide L , forcing $\deg L \geq 2^{|\sigma|} - 1$.

(2) By definition $\lambda_v \circ A = \lambda_{vA}$, with vA the row vector v times A . Extend the isomorphism between the lines $\mathbb{F}v$ and $\mathbb{F}e_{i_0}$ that sends v to e_{i_0} to a linear automorphism g of $\mathbb{F}^\sigma$, and let A be the transpose of the matrix of g . Then $vA = g(v) = e_{i_0}$. $\square$

3.3 The one-dimension step

Write the variable set as $\sigma \cup \{x_0\}$ and a point as (c, a') with $c \in \mathbb{F}_2$ and $a' \in \mathbb{F}_2^\sigma$. For $Q \in \mathbb{F}_2[x_0, x_\sigma]$ and $c \in \mathbb{F}_2$, let $Q(c, x') \in \mathbb{F}_2[x_\sigma]$ denote the substitution $x_0 = c$.

Lemma 3.4. *For all Q, c and a' , $\text{mult}_{a'}(Q(c, x')) \geq \text{mult}_{(c, a')}(Q)$ [Lean].*

Proof. The shift of $Q(c, x')$ at a' is $Q((c, a') + (w, z'))$ with $w = 0$. Its coefficients are the coefficients of $Q((c, a') + (w, z'))$ at monomials without w , and these have the same degree. $\square$

Theorem 3.5 (One-dimension step). *Let σ be a finite set, $n = |\sigma| + 1$ and $k < 2^n$. Let $P \in \mathbb{F}_2[x_0, x_\sigma]$ satisfy $\text{mult}_a(P) \geq k$ for every nonzero point a and $\text{mult}_0(P) < k$. Then there is $S \in \mathbb{F}_2[x_\sigma]$ with $\text{mult}_{a'}(S) \geq k$ for every nonzero a' , with $\text{mult}_0(S) = \text{mult}_0(P)$, and with*

$$\deg S + 1 \leq \deg P$$

[Lean].

In particular $D(n-1, k) \leq D(n, k) - 1$ for $n \geq 2$ and $k < 2^n$. Since the origin order is kept exactly, the same holds for the least degree with a prescribed origin order.

Proof. Let $\ell = \text{mult}_0(P) < k$ and let $L \neq 0$ be the homogeneous component of P of degree ℓ .

Choosing coordinates. We have $\deg L = \ell \leq k - 1 < 2^n - 1$, so Lemma 3.3 gives a nonzero form $\lambda_v \nmid L$ and an invertible A with $\lambda_v \circ A = x_0$. Put $Q = P \circ A$. By Lemma 3.2:

- Q satisfies the hypotheses, because A permutes the nonzero points;
- $\text{mult}_0(Q) = \ell$ and $\deg Q \leq \deg P$;
- the degree- ℓ component of Q is $L \circ A$;
- $x_0 \nmid L \circ A$: otherwise substituting A^{-1} would give $\lambda_v \mid L$.

The difference. Let $S = Q(0, x') + Q(1, x')$. Write $Q = \sum_{t \geq 0} x_0^t Q_t$ with $Q_t \in \mathbb{F}_2[x_\sigma]$. Then $S = \sum_{t \geq 1} Q_t$, since the $t = 0$ terms cancel in characteristic two.

- *Degree.* $\deg Q_t + t \leq \deg Q$, so $\deg S \leq \deg Q - 1$. Moreover $\deg Q \geq 1$, for otherwise $S = 0$, which the next item excludes.
- *Nonzero points.* For $a' \neq 0$, both $(0, a')$ and $(1, a')$ are nonzero, so Lemma 3.4 and Lemma 2.2(2) give $\text{mult}_{a'}(S) \geq k$.
- *Origin order, lower bound.* Lemma 3.4 gives $\text{mult}_0(Q(0, x')) \geq \ell$, and $\text{mult}_0(Q(1, x')) \geq k$ because $(1, 0) \neq 0$. So $\text{mult}_0(S) \geq \ell$.
- *Origin order, upper bound.* The constant term of $L \circ A$ in x_0 , namely $(L \circ A)(0, x')$, is nonzero because $x_0 \nmid L \circ A$. Take a monomial z^β with a nonzero coefficient in it. Then $|\beta| = \ell$, and the same coefficient appears in $Q(0, x')$. The coefficient of z^β in $Q(1, x')$ vanishes, because $\ell < k \leq \text{mult}_0(Q(1, x'))$. So z^β occurs in S , and $\text{mult}_0(S) \leq \ell$.

Finally $\deg S + 1 \leq \deg Q \leq \deg P$. □

3.4 The lower bound

Theorem 3.6. *Let $k \geq 1$ and $n \geq \lfloor \log_2 k \rfloor + 2$. Every k -admissible $P \in \mathbb{F}_2[x_1, \dots, x_n]$ satisfies*

$$\deg P \geq 2k + n - 2 - \lfloor \log_2 k \rfloor$$

[Lean].

Proof. Let $m = \lfloor \log_2 k \rfloor + 2$, so $k < 2^{m-1}$. Argue by induction on $n \geq m$.

For $n = m$, Proposition 3.1 gives $\deg P \geq 2k - \lfloor k/2^{m-1} \rfloor = 2k$, which is the claim.

For $n + 1 > m$, identify the variables with $\{x_0\} \cup \sigma$, where $|\sigma| = n$. This relabelling preserves degree, and by Lemma 2.2(5) it preserves multiplicities at the relabelled points. Since $k < 2^{m-1} \leq 2^{n+1}$, Theorem 3.5 gives a k -admissible S in n variables with $\deg S + 1 \leq \deg P$. The induction hypothesis bounds $\deg S$. □

4 The upper bound

4.1 Binary digit sums

Let $s_2(m)$ be the sum of the binary digits of $m \geq 0$.

Lemma 4.1. *For all $a, b, j, J \geq 0$:*

1. $s_2(a + b) \leq s_2(a) + s_2(b)$ and $s_2(2^j) = 1$. Hence a sum of r powers of two has digit sum at most r .
2. $s_2(a) \leq a$.
3. If $a \leq b$, then $2a + s_2(b) \leq 2b + s_2(a)$. That is, $a \mapsto 2a - s_2(a)$ is monotone.
4. $s_2(2^J - 1) = J$.

[Lean: parts 1–3] [Lean: part 4]

Proof. All four parts use the recursion $s_2(m) = (m \bmod 2) + s_2(\lfloor m/2 \rfloor)$.

1. The first inequality is proved by strong induction on $a + b$, using $\lfloor (a + b)/2 \rfloor = \lfloor a/2 \rfloor + \lfloor b/2 \rfloor + c$, where $c = \lfloor ((a \bmod 2) + (b \bmod 2))/2 \rfloor$ is the carry. The value $s_2(2^j) = 1$ follows from the recursion by induction on j .
2. This is standard; Mathlib proves it for every base.
3. Induct on $b - a$, using $s_2(b + 1) \leq s_2(b) + 1$, which is part 1 with second summand 1.
4. $2^{J+1} - 1$ is odd with $\lfloor (2^{J+1} - 1)/2 \rfloor = 2^J - 1$. □

4.2 The truncated product

Put $y_i = x_i^2 + x_i = x_i(1 + x_i)$.

Lemma 4.2. *In every commutative ring of characteristic two and for every t and $K \geq 0$, $\sum_{j < K} (t^2 + t)^{2^j} = t^{2^K} + t$. In particular, over $\mathbb{F}_2$,*

$$(1 + x) + \sum_{j < K} y^{2^j} = (1 + x)^{2^K}, \quad y = x^2 + x$$

[Lean].

Proof. By Frobenius, $(t^2 + t)^{2^j} = t^{2^{j+1}} + t^{2^j}$, so the sum telescopes. For the second form take $t = 1 + x$, for which $t^2 + t = x^2 + x$. □

Definition 4.3 (Truncated product). For $s \geq 1$, let $g_s \in \mathbb{F}_2[x_\sigma]$ be the sum, over all subsets $S \subseteq \sigma$ and all assignments $i \mapsto a_i$ of powers of two to $i \in S$ with $\sum_{i \in S} a_i \leq s - 1$, of

$$\prod_{i \in S} y_i^{a_i} \prod_{i \notin S} (1 + x_i)$$

[Lean].

In characteristic two the Catalan number C_{a-1} is odd exactly when a is a power of two. So g_s is the specialization to $\mathbb{F}_2$ of the Catalan truncation of Menezes [Men26, Section 3]. In Lean the pair (S, a) is encoded by a map $c : \sigma \rightarrow \{\perp, 0, \dots, s-1\}$: $c_i = \perp$ means $i \notin S$, and $c_i = j$ means $a_i = 2^j$. The sum runs over maps of cost $\sum_{c_i \neq \perp} 2^{c_i} \leq s-1$. The bound $j < s$ is automatic, because $2^j \leq s-1 < 2^s$. (The Lean statements hold for every $s \geq 0$ with natural-number subtraction, where $s-1 = 0$ for $s = 0$, so $g_0 = g_1$; only $s \geq 1$ is used.)

Theorem 4.4. *For every finite σ with $|\sigma| = n$ and every $s \geq 1$:*

1. $\text{mult}_a(g_s) \geq s$ for every nonzero $a \in \mathbb{F}_2^\sigma$;
2. $g_s(\mathbf{0}) = 1$;
3. $\deg g_s \leq n + 2(s-1) - s_2(s-1)$.

[Lean]

Proof. Expansion. By Lemma 4.2 with $K = s$,

$$\prod_{i \in \sigma} (1 + x_i)^{2^s} = \prod_{i \in \sigma} \left((1 + x_i) + \sum_{j < s} y_i^{2^j} \right).$$

Expanding the right-hand side gives the terms of Definition 4.3 of every cost. Hence the left-hand side equals $g_s + R$, where R collects the terms of cost at least s . This includes every term that uses a power $2^j > s-1$.

Multiplicity (1).

- Since $u^2 + u = 0$ on $\mathbb{F}_2$, each y_i vanishes at every point of $\mathbb{F}_2^\sigma$. Lemma 2.2(3) then gives each term of R multiplicity at least its cost, so at least s everywhere. By Lemma 2.2(2), so does R .
- At a nonzero a , choose i_0 with $a_{i_0} = 1$. Then $(1 + x_{i_0})^{2^s}$ has multiplicity at least $2^s \geq s$ at a , and so does the whole product.

In characteristic two, g_s equals the product plus R , so $\text{mult}_a(g_s) \geq s$.

Value (2). Every term with $S \neq \emptyset$ vanishes at $\mathbf{0}$, and the term with $S = \emptyset$ is $\prod(1 + 0) = 1$.

Degree (3). The term of (S, a) has degree at most $(n - |S|) + 2w$, where $w = \sum_{i \in S} a_i \leq s-1$. By Lemma 4.1(1), $|S| \geq s_2(w)$. By Lemma 4.1(3), $2w - s_2(w) \leq 2(s-1) - s_2(s-1)$. $\square$

4.3 Prescribing the origin order

Theorem 4.5. *Let σ be finite with $|\sigma| = n \geq 1$, fix $x_1 \in \sigma$, and let $0 \leq \ell < k$. The polynomial $P = y_1^\ell g_{k-\ell}$ satisfies:*

1. $\text{mult}_a(P) \geq k$ for every nonzero a ;
2. $\text{mult}_{\mathbf{0}}(P) = \ell$;
3. $\deg P \leq n + 2k - 2 - s_2(k - \ell - 1)$.

In particular, the least degree of a polynomial with multiplicity at least k off the origin and origin order exactly ℓ is at most $n + 2k - 2 - s_2(k - \ell - 1)$, in every dimension $n \geq 1$ [Lean].

Proof. 1. $\text{mult}_a(y_1^\ell) \geq \ell$ and $\text{mult}_a(g_{k-\ell}) \geq k - \ell$, so Lemma 2.2(3) gives the bound.

2. y_1 has origin order exactly one: its coefficient of x_1 is 1. Also $g_{k-\ell}(\mathbf{0}) = 1$ has origin order zero. Multiplicities add over $\mathbb{F}_2$ (Lemma 2.2(3)).
3. $\deg y_1^\ell \leq 2\ell$. Add the bound of Theorem 4.4(3) with $s = k - \ell \geq 1$. $\square$

Menezes proves this degree is optimal when $n \geq k - 1$ [Men26, Corollary 1.3]. For $n < k - 1$ we prove optimality only for the orders that attain $D(n, k)$ (Section 5).

5 Proof of the main theorem

Theorem 5.1 (The formula below 2^{n-1}). *Let $k \geq 1$.*

1. *For every $n \geq \lfloor \log_2 k \rfloor + 2$, every k -admissible $P \in \mathbb{F}_2[x_1, \dots, x_n]$ has $\deg P \geq 2k + n - 2 - \lfloor \log_2 k \rfloor$.*
2. *For every $n \geq 1$, some k -admissible $P \in \mathbb{F}_2[x_1, \dots, x_n]$ has $\deg P \leq 2k + n - 2 - \lfloor \log_2 k \rfloor$.*

Hence $D(n, k) = 2k + n - 2 - \lfloor \log_2 k \rfloor$ whenever $n \geq \lfloor \log_2 k \rfloor + 2$, equivalently $k < 2^{n-1}$, and a polynomial of exactly this degree exists [Lean].

Proof. Part 1 is Theorem 3.6.

For part 2, let $J = \lfloor \log_2 k \rfloor$, so $2^J \leq k$, and let $\ell = k - 2^J < k$. The polynomial of Theorem 4.5 is k -admissible with origin order ℓ , and its degree is at most

$$n + 2k - 2 - s_2(2^J - 1) = n + 2k - 2 - J$$

by Lemma 4.1(4).

For $k \geq 1$, $k < 2^{n-1}$ is equivalent to $\lfloor \log_2 k \rfloor < n - 1$, which is $n \geq \lfloor \log_2 k \rfloor + 2$. $\square$

Corollary 5.2. *For $n \geq 1$ and $1 \leq k < 3 \cdot 2^{n-1}$, $D(n, k) = 2k + n - 2 - \lfloor \log_2 k \rfloor$ [Lean].*

Proof. For $k < 2^{n-1}$ this is Theorem 5.1. The upper bound holds in every dimension, so it remains to match it with Proposition 3.1 in two cases.

- If $2^{n-1} \leq k < 2^n$, then $\lfloor \log_2 k \rfloor = n - 1$ and $\lfloor k/2^{n-1} \rfloor = 1$, so both bounds equal $2k - 1$.
- If $2^n \leq k < 3 \cdot 2^{n-1}$, then $\lfloor \log_2 k \rfloor = n$ and $\lfloor k/2^{n-1} \rfloor = 2$, so both equal $2k - 2$. $\square$

For $k \geq 2^{n-1}$ the value comes from the hyperplane covers of [BDDM23].

Theorem 5.3 (Hyperplane covers [BDDM23, Theorem 1.2(a) with $d = 1$, arXiv v1]). *Let $n \geq 1$ and $k \geq 2^{n-1}$, and write $k = a2^{n-1} + b$ with $0 \leq b < 2^{n-1}$. Then*

$$P = \prod_{u \in \mathbb{F}_2^n \setminus \{\mathbf{0}\}} (1 + u \cdot x)^a \cdot (x_1^2 + x_1)^b$$

is k -admissible and $\deg P \leq 2k - \lfloor k/2^{n-1} \rfloor$ [Lean].

In the language of [BDDM23], P is the product of the linear forms of a cover by a copies of the $2^n - 1$ hyperplanes $u \cdot x = 1$, which avoid the origin, and b copies of the parallel pair $x_1 = 0, x_1 = 1$. Their theorem gives the least size of a cover for every $k \geq 2^{n-2}$; we use only this upper bound for $k \geq 2^{n-1}$, in the polynomial form above, which is what the Lean file proves.

Proof of Theorem 1.2. For $k < 2^{n-1}$ the value is Theorem 5.1. For $k \geq 2^{n-1}$, Proposition 3.1 gives the lower bound and Theorem 5.3 the upper bound. By Proposition 3.1 every admissible polynomial has degree at least $2k - \lfloor k/2^{n-1} \rfloor$, so for the equivalence it remains to compare $D(n, k)$ with this bound in three cases.

- If $k \geq 2^{n-1}$, they are equal by the first part.
- If $2^{n-2} \leq k < 2^{n-1}$, then $\lfloor \log_2 k \rfloor = n - 2$ and $\lfloor k/2^{n-1} \rfloor = 0$, so both equal $2k$.
- If $k < 2^{n-2}$, then $n \geq 3$, $\lfloor \log_2 k \rfloor \leq n - 3$ and $\lfloor k/2^{n-1} \rfloor = 0$, so $D(n, k) \geq 2k + 1$.

The upper bound in every dimension is Theorem 5.1(2). □

At $k = 3 \cdot 2^{n-1}$ the logarithmic expression is $2k - 2$, while $D(n, k) = 2k - 3$ by Theorem 1.2.

Remark 5.4 (Which origin orders are extremal). Let $k < 3 \cdot 2^{n-1}$, so that $D(n, k) = 2k + n - 2 - \lfloor \log_2 k \rfloor$. Theorem 4.5 gives degree $n + 2k - 2 - s_2(k - \ell - 1)$ at origin order ℓ . This equals $D(n, k)$ exactly when $s_2(k - \ell - 1) = \lfloor \log_2 k \rfloor$, for instance $\ell = k - 2^{\lfloor \log_2 k \rfloor}$. The 69 per-order minima computed exactly for $n \leq 7$ in the [research notebook](#) all equal $n + 2k - 2 - s_2(k - \ell - 1)$, including 15 values with $n < k - 1$. That computation is evidence, not part of the formalized result.

What does not follow

- **Per-order minima below $n = k - 1$.** For $n < k - 1$ we do not prove that $n + 2k - 2 - s_2(k - \ell - 1)$ is the least degree at an origin order ℓ with $s_2(k - \ell - 1) < \lfloor \log_2 k \rfloor$. For $n \geq k - 1$ this is part of Menezes' theorem [Men26].
- **Other fields and grids.** The lower bound uses the binary setting in several places: the cancellation of the x_0 -free part in $Q(0, x') + Q(1, x')$, the finite number of linear forms in Lemma 3.3, and $S = \mathbb{F}_2$ in Proposition 3.1. The upper bound uses characteristic two in the telescoping identity. We make no claim about other fields or grids.

6 Evidence and the research framework

6.1 Development and checks

The results were developed in a public research notebook [Notebook]. Its dated entries record the exact computations, the failed routes and the reviews.

- **Exact computation.** Before the general statement was proved, minimum degrees were computed exactly for $n \leq 7$ by rank computations over $\mathbb{F}_2$. This produced the formula.
- **Earlier upper bound.** Until the literature search found Menezes' construction, the upper bound was proved only for $k \leq 31$, by explicit symmetric factors found by a solver and checked independently.
- **Checks of the construction.** The construction of Theorem 4.5 was checked by exact multiplicity computation in 417 cases with $k \leq 12$ and $n \leq 7$, 305 of them below Menezes' range $n \geq k - 1$.
- **Review.** The informal proofs of the step, the lower bound and the construction were checked by fresh-context reviewers before being recorded.

- **Formalization.** The whole chain of Theorem 1.2 was then formalized. The Lean proofs follow the arguments given here.

The one-dimension step, the truncated product and the Schwartz–Zippel lemma use generalized forms, such as every $n \geq 1$, every s and every field; this paper states them in those forms.

6.2 Noemesis

The work used *Noemesis*, an open framework for autonomous mathematical research, distributed with the repository [Repo]. It consists of instructions, tools and record-keeping conventions that let an AI assistant continue an investigation across sessions with inspectable evidence:

- a living overview and a claim index;
- dated, append-only research entries, including failures;
- protected and timed computations with retained outputs;
- dependency-indexed Lean formalization, linked from the claim index.

These mechanisms make the work recoverable and reviewable. They do not make a recorded proof correct; the Lean formalization does that for the statements carrying [Lean] links.

6.3 Contributions and AI assistance

- **The earlier note** [Note]. Proposition 3.1 was drafted with GPT-6 Astra in the Codex agent.
- **This paper.** The rest of the mathematics was produced by Claude Opus 5.5 in Claude Code, working inside the framework above under the author’s direction: the exact computations, the formula, the one-dimension step, the literature search that found [Men26], the characteristic-two proof of the construction, the complete Lean formalization, and the draft of this paper. Fresh-context Claude Opus reviewers checked the informal arguments.
- **The author.** The author chose the problem, directed the investigation and designed the research framework.

The paper has not been externally peer reviewed. Formal verification covers the statements with [Lean] links, under the definitions in the linked files. It does not cover the prose or the comparison with the literature.

The construction of Section 4 is due to Menezes [Men26], and the multiplicity Schwartz–Zippel lemma to Dvir, Kopparty, Saraf and Sudan [DKSS13]. Original manuscript material is distributed under the [Creative Commons Attribution 4.0 International license \(CC BY 4.0\)](#).

A Formal-verification map and reproduction

The arguments are complete in the body; reading them requires no Lean. This appendix locates the formal declarations and gives commands to check them. File paths are relative to `formalization/`. The table uses the prefix `MR.` for `MathResearch.`; the external inputs lie under `MathResearch.ThirdParty`.

Paper statement	Lean file and declarations
Hasse multiplicity, Definition 2.1; Lemma 2.2	claims/HasseMultiplicity.lean MR.HasseMultiplicity.mult, coe_le_mult_iff, min_le_mult_add, add_le_mult_mul, mult_mul, mult_eq_zero_iff, mult_rename_equiv
Multiplicity Schwartz–Zippel [DKSS13], Theorem 2.3	third-party-claims/MultiplicitySchwartzZippel.lean MR.ThirdParty.MultiplicitySchwartzZippel.multiplicity_ schwartz_zippel
Schwartz–Zippel bound over $\mathbb{F}_2$, Proposition 3.1	claims/BinaryMultiplicityCoverBound.lean MR.BinaryMultiplicityCoverBound.cover_bound, cover_bound_of_origin
Linear substitutions, Lemma 3.2; restriction, Lemma 3.4	claims/OneDimensionStep.lean MR.OneDimensionStep.homogeneousComponent_subst, totalDegree_subst_le, subst_comp, mult_zero_subst, mult_subst, mult_restrict
Avoiding a linear form, Lemma 3.3	claims/LinearFormAvoidance.lean MR.LinearFormAvoidance.exists_linForm_not_dvd, exists_subst_linForm_eq_X
One-dimension step, Theorem 3.5	claims/OneDimensionStep.lean MR.OneDimensionStep.one_dimension_step
Lower bound, Theorem 3.6	claims/BinaryMultiplicityLowerBound.lean MR.BinaryMultiplicityLowerBound.lower_bound
Digit sums, Lemma 4.1	claims/TruncatedProduct.lean MR.TruncatedProduct.s2_add_le, s2_two_pow, s2_le_self, two_mul_add_s2_le; MR.BinaryMultiplicityDegreeFormula.s2_two_pow_sub_one
Telescoping, Lemma 4.2; truncated product, Theorem 4.4	claims/TruncatedProduct.lean MR.TruncatedProduct.telescope, g, truncated_product
Per-order construction, Theorem 4.5	claims/PerOrderConstruction.lean MR.PerOrderConstruction.per_order_construction
Formula below 2^{n-1} , Theorem 5.1; upper bound of Theorem 1.2; Definition 1.1	claims/BinaryMultiplicityDegreeFormula.lean MR.BinaryMultiplicityDegreeFormula.Admissible, degree_formula, degree_formula_exact, degree_formula_of_lt_two_pow
Extended range, Corollary 5.2	claims/BinaryMultiplicityDegreeExtendedRange.lean MR.BinaryMultiplicityDegreeExtendedRange.degree_formula_ extended
Hyperplane covers [BBDM23], Theorem 5.3	third-party-claims/HyperplaneCoverUpperBound.lean MR.ThirdParty.HyperplaneCoverUpperBound.hyperplane_cover_ upper_bound, card_dot_eq_one
Main theorem, Theorem 1.2	claims/BinaryMultiplicityDegreeComplete.lean MR.BinaryMultiplicityDegreeComplete.degreeMin, degree_complete, cover_bound_attained_iff

Reading the formal statements. In Lean, points are functions $\text{Fin } n \rightarrow \text{ZMod } 2$, degree is `MvPolynomial.totalDegree`, and multiplicity is `MR.HasseMultiplicity.mult`, the order of $P(a+z)$ in $\mathbb{N}_\infty$. The one-dimension step is stated for the variables Option σ , with none playing the role of

x_0 . The lower bound relabels $\text{Fin}(n + 1)$ as $\text{Option}(\text{Fin } n)$.

Commands. The project pins Lean 4.34.0-rc2 and Mathlib in `lakefile.lean` and `lake-manifest.json`. With [elan](#), Git and Python 3 installed:

```
git clone https://github.com/kbr-/math-research.git bmd-check
cd bmd-check
git checkout --detach 8651228097422970d147972353bbf9a1c6ed6938
cd formalization
export MATHLIB_NO_CACHE_ON_UPDATE=1
lake exe cache get claims/BinaryMultiplicityDegreeComplete.lean
python3 verify.py --target claims/BinaryMultiplicityDegreeComplete.lean \
  --out verification.txt
```

The target imports every module in the table, so the build covers the whole chain, and its axiom report is transitive. The verifier builds the proofs, prints each listed declaration’s type, and rejects unfinished proofs and nonstandard axioms. The expected axiom report for the main theorem is

```
'MathResearch.BinaryMultiplicityDegreeComplete.
degree_complete' depends on axioms:
[propxt, Classical.choice, Quot.sound]
```

These are Lean’s standard foundations.

References

- [AF93] Noga Alon and Zoltán Füredi. Covering the cube by affine hyperplanes. *European Journal of Combinatorics* **14**(2) (1993), 79–83. [doi:10.1006/eujc.1993.1011](#).
- [BBDM23] Anurag Bishnoi, Simona Boyadzhiyska, Shagnik Das and Tamás Mészáros. Subspace coverings with multiplicities. *Combinatorics, Probability and Computing* **32**(5) (2023), 782–795. [doi:10.1017/S0963548323000123](#). Question 4.3 appears as Question 5.2 in [arXiv:2101.11947v1](#).
- [DKSS13] Zeev Dvir, Swastik Kopparty, Shubhangi Saraf and Madhu Sudan. Extensions to the method of multiplicities, with applications to Kakeya sets and mergers. *SIAM Journal on Computing* **42**(6) (2013), 2305–2328. [doi:10.1137/100783704](#). Lemma 2.7 is Lemma 8 of [arXiv:0901.2529v2](#).
- [Men26] Dean Menezes. Characteristic drops for high-order vanishing on the hypercube. [arXiv:2609.19009v1](#), 14 August 2026.
- [Note] Kamil Braun. A note on a question about polynomial multiplicities over $\mathbb{F}_2$. Preprint, version 1, 20 September 2026. <https://github.com/kbr-/math-research/tree/main/publications/binary-polynomial-multiplicity>.
- [Notebook] Kamil Braun, with the AI assistance described in Section 6.3. Binary multiplicity degree: research notebook, 2026. <https://kbr.is-a.dev/math-research/branches/binary-multiplicity-degree/>.
- [Repo] Kamil Braun, with the AI assistance described in this manuscript. *Noemesis: A Self-Improving Framework for Autonomous Mathematical Research*: source repository and live research notebook, 2026. <https://github.com/kbr-/math-research>. Version containing the formal proofs cited here: Git commit [Obb1f410](#), 24 September 2026.
- [SW20] Lisa Sauermann and Yuval Wigderson. Polynomials that vanish to high order on most of the hypercube. [arXiv:2010.00077v2](#), 28 March 2022.